

involving computers, the internet, and video conferencing enable us to interact with people often sitting at different geographic locations easily. In the concurrent engineering approach, the team will discuss and evolve design ideas on a real-time basis. In this process a large number of iterations are possible.

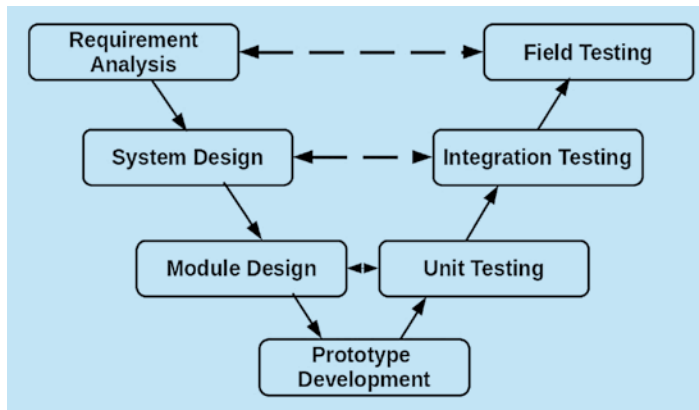


Fig. 3: Vee model

Spiral model

The spiral development model is one of the simplest concurrent engineering approaches (see Fig. 2). In this model, development happens through many development workshops or sprints. These are intense and focused interactions involving all team members to achieve a specific goal. There is no rule on the number of sprints in the spiral process. During the sprint, the team will go through a cycle of planning, developing, evaluating, and suggesting further improvements.

The product development plan starts with a sketchy concept. It gradually evolves into a proof of concept mock-up, a working prototype, pilot samples for test marketing, and finally the marketable product.

The spiral development model enables rapid product development. It uses rapid prototype development technologies like 3D printing, CAD-CAM, simulation, etc. In the spiral development model, all team members can work together.

For instance, while the designers are making a circuit design, the supply chain team can use the shared bill of materials to develop suppliers. The costing team can work out the production economics, while the production team can plan their production strategy right from the start of designing. Each of these teams

gives their feedback and works out the best possible options.

Vee (V) model

A spiral development model faces the challenge of keeping the focus on the original requirement. As the project becomes large, the team tends to lose focus. We also have the challenge of engaging a wide-spread, complex team for fruitful discussions. In 1979, Barry W. Boehm proposed another model primarily for software development, which is now known as the Vee model. For product development Vee model is quite popular.

The Vee model is an extension of the traditional waterfall model. In this, the distinct phases of the waterfall model and the spiral model are superimposed. This results in distinct phases of conceptualisation, development, and validation. In the Vee model validation phases are tied-up with conceptualisation phases (see Fig. 3).

The structure of the Vee model enables us to engage smaller and focused teams at different phases of product development. Unlike the spiral model, Vee model phases have a specific and well-defined deliverable. This structure helps to manage the project, track progress, and keep the team focused on main requirements. Unlike the spiral model, the author has worked with

a development team that comprised around seventy members spread across five continents successfully using the Vee model.

While popular, the Vee model is criticised for being too rigid. Where requirements are fuzzy and contain a high risk of changing, the Vee model is not suitable. In such a situation, it is often advisable to rapidly get

a small product lot in the market and gradually evolve the product in continuous new version releases. We see this practice in operation in high technology products like cell phones, operating systems, computers, etc.

Apart from these, there are many other development models. We have to select the correct development model for the product. No model will fit all sorts of products. Each of them has certain advantages, and each of them has certain disadvantages. However, the popular models described here will fit the most common product development requirements.

The waterfall model is good for the development of our charge controller. It is a small and simple product and will not require a large team or much development effort. But we shall use the Vee development model to demonstrate the process.

Requirements for CE

Concurrent engineering processes need some basic infrastructures. First of all, there has to be participation from all functional departments. The development team should have dedicated members who are knowledgeable and empowered to take decisions. Preferably. The development team should sit together. If that is not possible, they must meet in a well-defined periodic manner and devote specific time to the project.